

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

ST. ANDREW'S JUNIOR COLLEGE
2016 JC2 Preliminary Examinations

H2 BIOLOGY**9648/2****Paper 2: Core**

Monday

29th August 2016

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question only.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/9
3	/13
4	/7
5	/11
6	/17
7	/11
Total	/80

This document consists of **19** printed pages.

[Turn over

Section A

Answer **all** questions.

1. Lysosomes play an important role in autophagy as they contain enzyme complexes that can break down worn out organelles. A section of the lysosomal membrane is magnified and shown in Fig. 1.1.

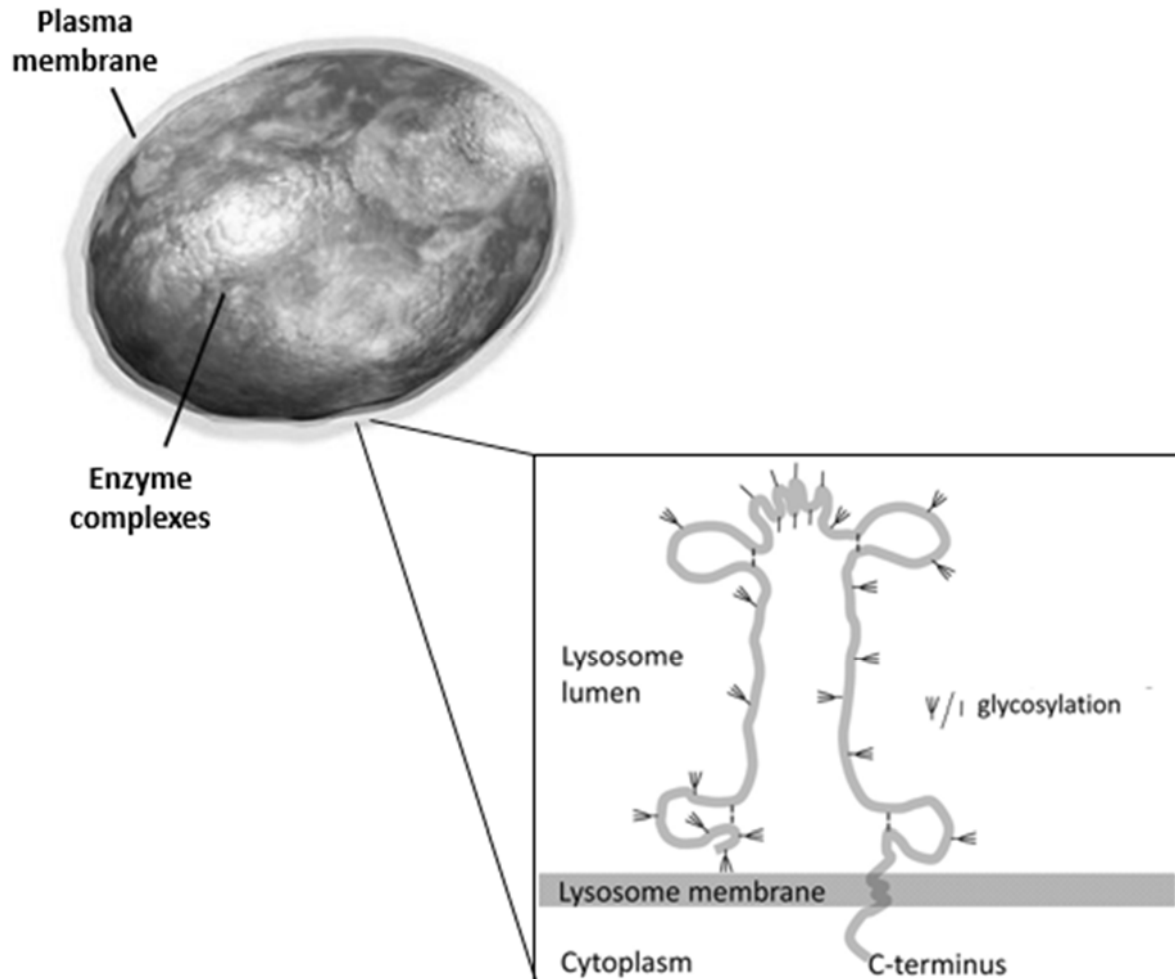


Fig. 1.1

(a) Describe and explain how the structure of the lysosomal membrane protects it from being degraded by the hydrolytic enzyme complexes it contains.

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.....[2]

(b) Collagen is an important structural component of skin, ligament and tendons. Describe the various levels of protein folding that gives it its fibrous structure.

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.....[4]

(c) Fig. 1.2 and 1.3 show the structures of lysine and glycine respectively. In the space provided, draw a lys-gly dipeptide and label the peptide bond.

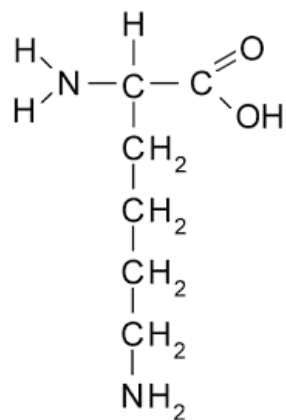


Fig. 1.2

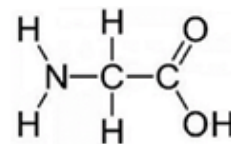


Fig. 1.3

.....[1]

(d) Fig. 1.4 shows some onion cells undergoing mitosis.

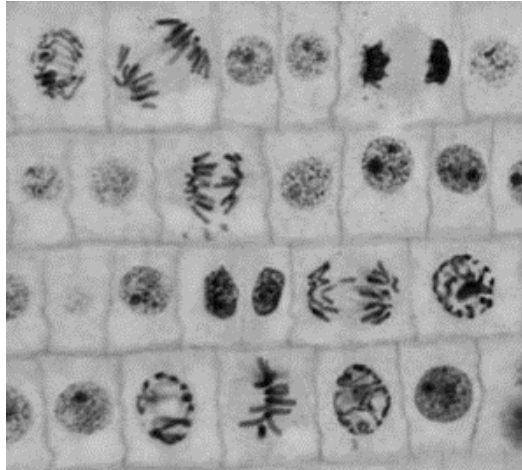


Fig. 1.4

(i) Identify (with arrows and labels) one cell each that is undergoing prophase, metaphase, anaphase and telophase.

.....[1]

(ii) Describe the events of anaphase.

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[2]

(iii) Describe the process of cytokinesis in the onion cells.

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[2]

[Q1 Total: 12]

2. Fig. 2.1 shows a process happening in a eukaryotic cell.

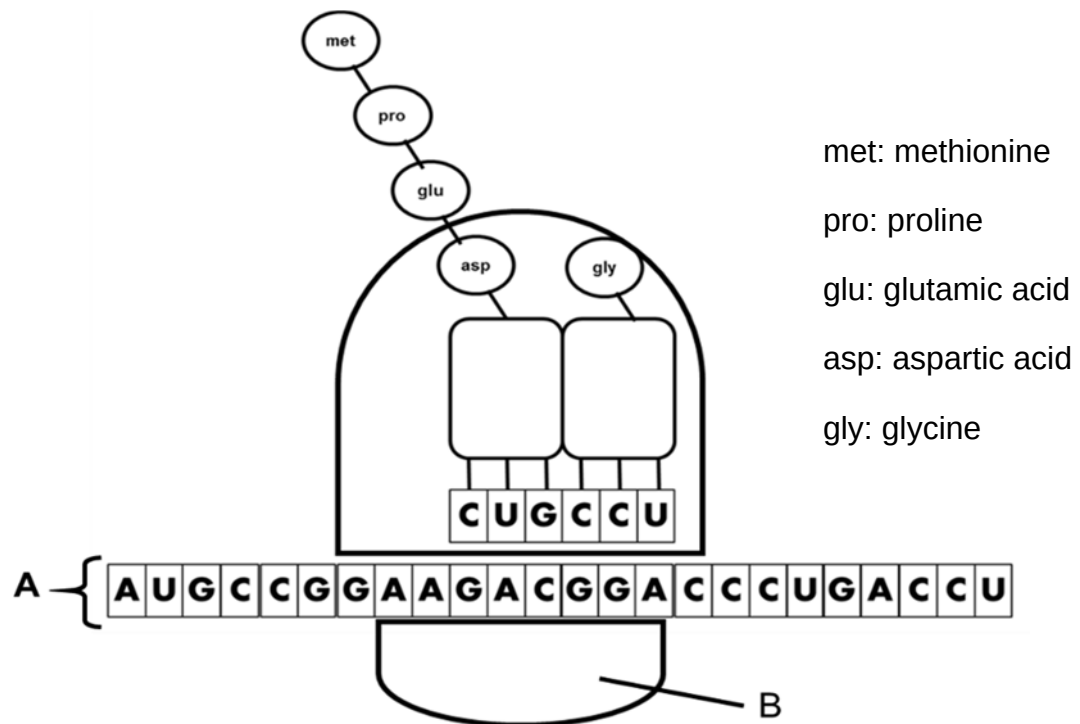


Fig. 2.1

(a) Name the structures labelled **A** and **B**. [1]

A:

B:

(b) Relate the structure of **A** to its role in protein synthesis.
.....
.....[1]

(c) Outline one change to **A**, following completion of transcription in the nucleus before it can be used for the process in Fig. 2.1.
.....
.....[1]

(d) How many amino acids does the protein, encoded by the section of **A** shown in Fig. 2.1, have?
.....[1]

(e) An antibiotic which affects the elongation stage of the process in Fig. 2.1, is added shortly after initiation, such that truncated polypeptides are formed instead.

With reference to Fig. 2.1, suggest and explain how the antibiotic works to give rise to a tripeptide.

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.....[3]

(f) Tetracycline is an example of a ribosome-targeting antibiotic which is effective only towards bacterial cells.

Suggest why tetracycline has no effect on eukaryotes.

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.....[2]

3. A study published in the *Proceedings of the National Academy of Sciences* revealed a plausible strategy to facilitate the replacement of antibiotic resistant pathogens with sensitive ones in hospitals, which faced a higher rate of evolution of antibiotic resistance.

Researchers developed and administered two different phages that target the common gut microbe *E. coli*. The first was just a standard “lytic” phage that results in the death of the bacterial cell. The second phage, called a “temperate” phage, was a little more special. The researchers engineered it to possess a well-known gene-editing system that, when injected into the host cell, both removes their antibiotic resistance genes and renders them resistant to the lytic phage at the same time.

To select for bacteria that has been successfully edited by the temperate phage, the lytic phage is added. Lytic phages only target bacteria that has not undergone gene-editing, thus selecting for successfully edited bacteria that are now antibiotic-sensitive.

Both phages designed according to this strategy are used on hospital surfaces and hand sanitizers to restore a “healthy balance” in the population of antibiotic-sensitive bacteria and antibiotic-resistant bacteria.

(a)(i) Explain how the structure of a “lytic” phage facilitates the “death of the bacterial cell.”

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[2]

(ii) Suggest why it is important to restore a “healthy balance” in the population of antibiotic-sensitive bacteria and antibiotic-resistant bacteria in hospitals.

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[1]

(iii) Contrast the life cycle of the “temperate” phage and life cycle of a HIV.

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[1]

.....[4]

Suggest why homologous recombination with the host cell genome is possible in generalised transduction.

.....[1]

trpR⁺*_trpP*⁺*_trpO*⁻*_trpE*⁺*_trpD*⁺*_trpC*⁺*_trpB*⁺*_trpA*⁺

Explain how this bacteria strain will respond in the presence of tryptophan (+: wild type allele; - : loss of function allele).

[4]

[Q4 Total: 7]

5. Coat colour in mice is determined by two gene loci, **B/b** and **D/d**.

The alleles give the following phenotypes:

B black **b** brown
D coloured **d** albino (white)

A cross between two black coat mice that are heterozygous at both gene loci produces F₁ offspring of the following phenotypes:

Black coat	675
Brown coat	225
Albino coat	300

(a) Define the term heterozygous.

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[1]

(b) Draw a genetic diagram to show the cross described.

[4]

(c) Explain how the interaction between the two genes B and D gives rise to albino coat colour.

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.....[4]

A wheat breeder crossed two cultivars of wheat that varied in kernel colour – one with dark red kernel, another with white kernel (a kernel is the seed from which the wheat plant grows)

He noted that the F_1 generation was intermediate in colour (light red), but the F_2 generation ranges in colours from white to dark red.

Fig. 5.1 shows the range of kernel colour in the F_2 generation.

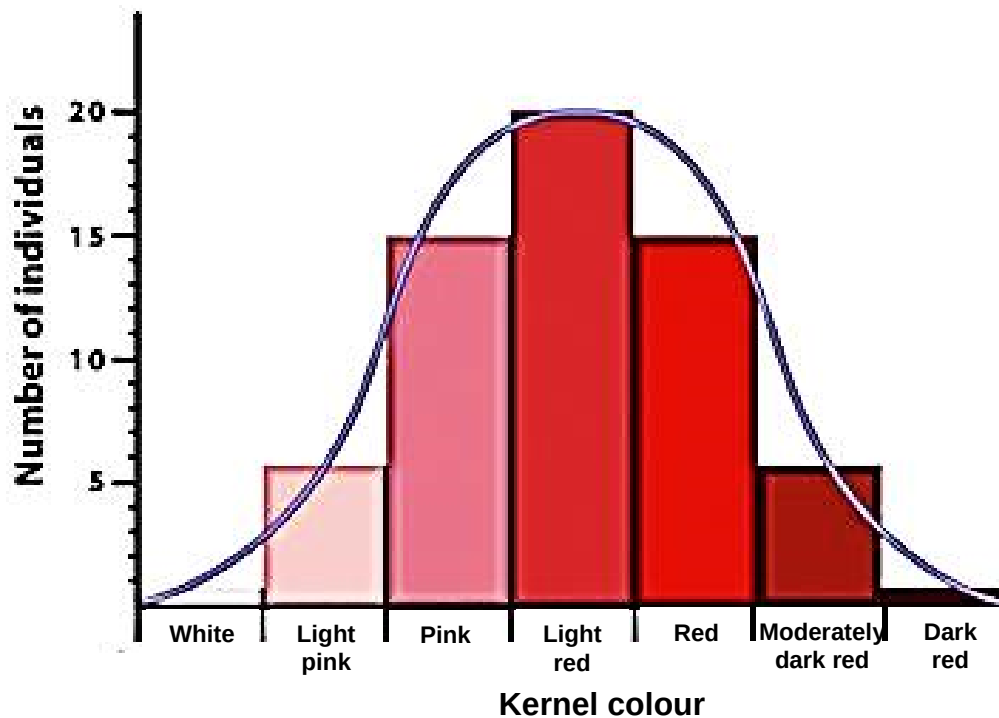


Fig. 5.1

(d) Explain the genetic basis behind the wide range of kernel colours.

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6. It has been hypothesized that the neurotransmitter, serotonin, plays a part in appetite control. A consistent pattern of reduction in hunger motivation is seen in human studies with a variety of drugs which induce the release of serotonin.

Patients receiving these drugs report both a lower frequency and a reduced strength of urges to eat, together with the feeling of being more in control of their eating. Such serotonergic drugs, such as Dexfenfluramine, can exert a continued suppression of appetite to bring about subsequent weight loss.

Fig 6.1 shows the diagram of a synapse at which serotonin is the neurotransmitter.

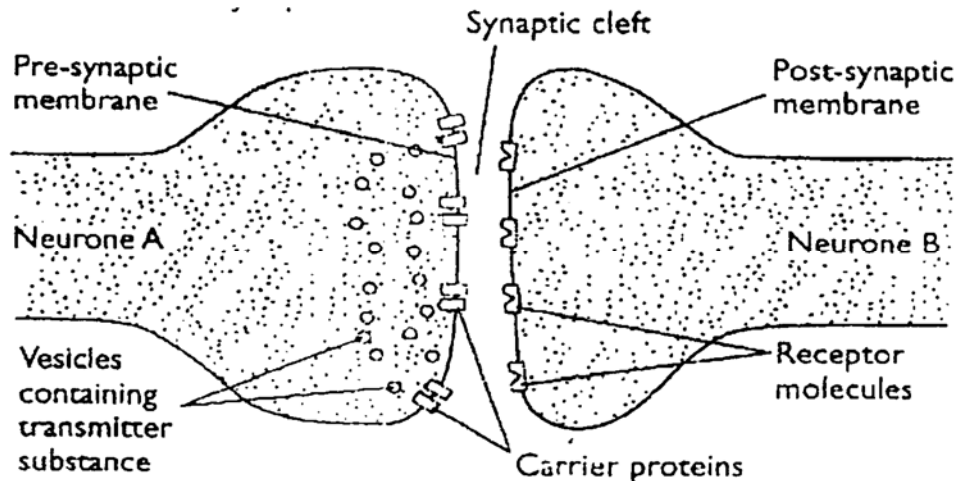


Fig 6.1

(a)(i) Describe how an impulse is initiated in Neurone B and transmitted to the brain, after the ingestion of Dexfenfluramine, to achieve the perception of fullness.

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Serotonin is normally rapidly reabsorbed from the synaptic cleft by carrier proteins in the pre-synaptic membrane.

(ii) Suggest one advantage of rapidly reabsorbing serotonin.

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.....[1]

(iii) Explain the effect of mitochondrial poisons, which inhibit cytochromes/electron carriers, on synaptic transmission.

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.....[2]

Serotonin receptors consist of several subtypes. In addition to ligand-gated ion channels, another subtype of serotonin receptors is shown in Fig 6.2, which participates in a transduction pathway that reduces the generation of beta amyloid, a compound involved in Alzheimer's disease. Raf, MEK and ERK are kinases.

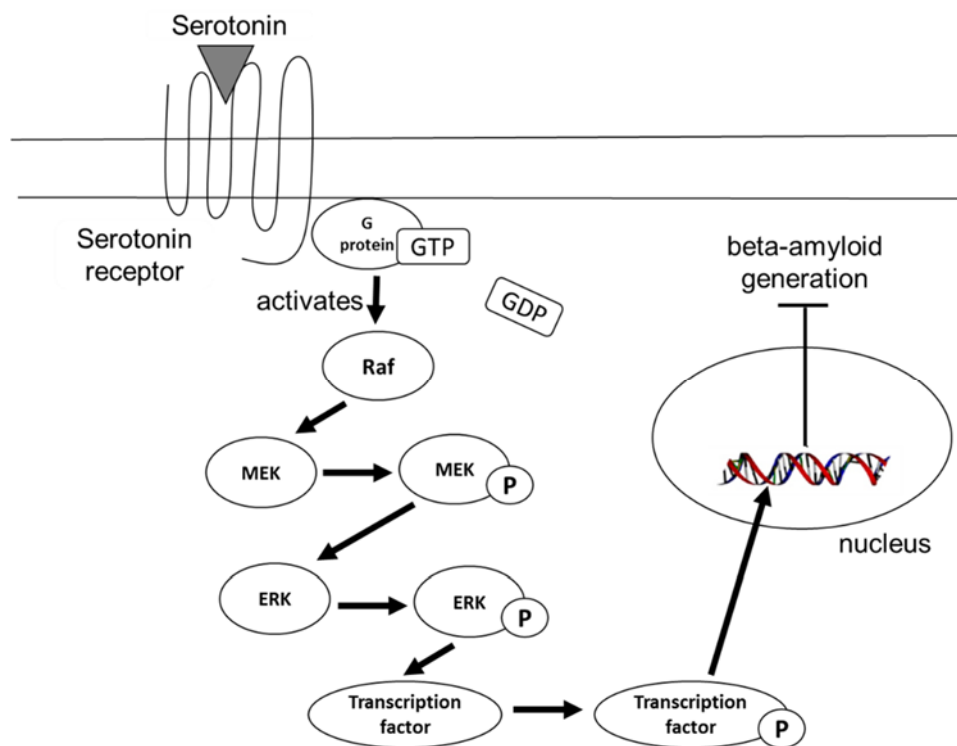


Fig. 6.2

(b)(i) Explain how the receptor in Fig.6.2 stays embedded in the plasma membrane and retains its mobility in the membrane.

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.....[3]

(ii) With reference to Fig. 6.2, explain how the binding of serotonin leads to a reduction in beta-amyloid generation.

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(c) There are reactions in plants which are crucial for its survival and growth. Explain the effect of a base addition to the gene coding for RuBisCO (Ribulose-1,5-bisphosphate carboxylase) enzyme.

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.....[3]

7. Batesian mimicry is a type of mimicry in which a harmless organism is protected by its resemblance to another species which is avoided by predators. An example of this is the king snake, which is quite harmless. However, it has a similar colour pattern as the coral snake which is highly venomous. As a result, predators of snakes that avoid the coral snake also avoid the king snake as well.

(a)(i) Suggest an evolutionary advantage to the king snake for merely resembling a venomous coral snake instead of producing true venom.

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.....[1]

(ii) Discuss how natural selection could have led to the evolution of the colour pattern in the king snake.

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.....[4]

(b) DNA–DNA hybridisation is a technique used to determine the similarity of DNA from different species.

In this technique:

- DNA is extracted from four king snake species and cut into pieces using special enzymes;
- the DNA is heated to separate the double strands into single strands;
- single-stranded DNA from different pairs of king snake species are mixed together and cooled so that double strands of hybrid DNA form.

Fig. 7.1 shows the temperature (°C) needed to separate the hybrid DNA strands of the four king snake species, *L. getula*, *L. calligaster*, *L. triangulum* and *L. nigrita*.

Fig. 7.1

	<i>L. getula</i>	<i>L. calligaster</i>	<i>L. triangulum</i>	<i>L. nigrita</i>
<i>L. getula</i>	91	84	89	80
<i>L. calligaster</i>		90	86	87
<i>L. triangulum</i>			92	82
<i>L. nigrita</i>				90

(i) Describe and explain the difference in denaturation temperature for the *L. getula*–*L. triangulum* hybrid DNA and the *L. getula*–*L. nigrita* hybrid DNA.

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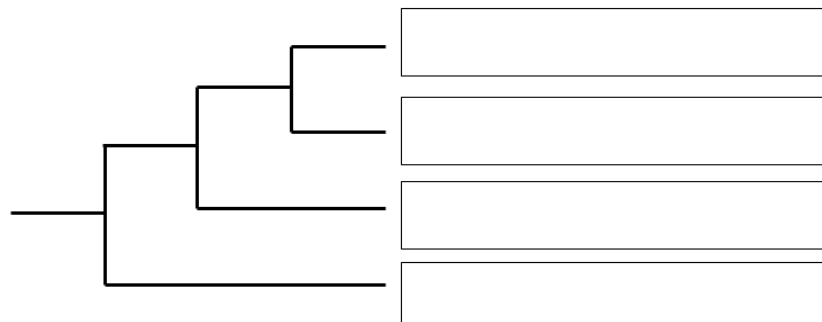
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(ii) Using the data in Fig. 7.1, fill in phylogenetic diagram below to show the evolutionary relationship between the four species of king snakes.

.....[1]



(iii) Discuss how the neutral theory of evolution could be used to determine when two species of king snakes diverged from their common ancestor.

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.....[2]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Explain the advantages and significance of cell signalling. [5]
- (b)** Secretion of glucagon plays a critical role the regulation of glucose concentration in the bloodstream.
- With reference to the stages of cell signalling, describe how glucagon performs this role. [10]
- (c)** Discuss the differences between the cell signalling pathways of insulin and glucagon. [5]

[Total: 20]

OR

- 9** **(a)** Describe, using examples, how homology supports Darwin's theory of natural selection. [10]
- (b)** Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [10]

[Total: 20]